

Tianyue H. Zhang

Email: tianyue.zhang@mila.quebec · Website: tianyuehz.github.io

Github: [tianyuehz](https://github.com/tianyuehz) · Publication: [Google Scholar](#)

Education

Sept. 2023 - Present	Mila/Université de Montréal GPA: $A^+ \equiv 4.33$ <i>Ph.D. Computer Science, DIRO</i> Supervised by Dr. Simon Lacoste-Julien
Sept. 2021 - Sept. 2023	University of British Columbia GPA: $95.3/100 \equiv 4.33$ <i>M.Sc. Computer Science</i> Supervised by Dr. Mark Schmidt
Sept. 2015 - Sept. 2019	University of British Columbia GPA: $85/100 \equiv 3.95$ <i>B.Sc. Combined Honours in Computer Science and Mathematics</i> Dean's Honour List; Graduated with Distinction

Publications

Signal Frequency Imbalance and Ill-Conditioning [Paper](#)

HiLD workshop at ICML 2026

T. Zhang, F. Bach, F. Kunstner

Deriving Hyperparameter Scaling Laws via Modern Optimization Theory [Paper](#)

Sci4DL workshop at ICLR 2026

E. Shulgin, D. von Rütte, T. Zhang, N. Ajroldi, B. Schölkopf, A. Orvieto

Understanding Adam Requires Better Rotation Dependent Assumptions [Paper](#)

NeurIPS 2025

T. Zhang^{*}, L. Maes^{*}, A. Milligan, A. Jolicoeur-Martineau, I. Mitliagkas, D. Scieur, S. Lacoste-Julien, C. Guille-Escuret

Addressing Concept Mislabeling in Concept Bottleneck Models Through Preference Optimization [Paper](#)

ICML 2025; Bi-Align Workshop Workshop at ICLR 2025 (Oral)

E. Penalzoza^{*}, T. Zhang, L. Charlin, M. E. Zarlenga

Connecting Thompson Sampling and UCB: Towards More Efficient Trade-offs Between Privacy and Regret [Paper](#)

ICML 2025

B. Hu, Z. Huang, T. Zhang, M. Lecuyer, N. Hegde

On PI Controllers for Updating Lagrange Multipliers in Constrained Optimization [Paper](#)

ICML 2024

M. Sohrabi^{*}, J. Ramirez^{*}, T. Zhang, S. Lacoste-Julien, J. Gallego-Posada

Optimistic Thompson Sampling for Episodic Reinforcement Learning [Paper](#)

UAI 2023

B. Hu, T. Zhang, N. Hegde, M. Schmidt

Thesis

Optimistic Thompson Sampling: Strategic Exploration in Bandits and RL [Thesis](#)

Supervisory committee: Dr. Mark Schmidt and Dr. Mathias Lecuyer

Selected Awards

2025	NSERC PGSD, \$40000 annually
2025	Mila Excellence Scholarships – Women in AI, \$8000 annually
2023	Professor Cho Diversity Scholarship, \$1500
2022	UBC Cloud Innovation Center Fellowship, \$20000
2019	NSERC Undergraduate Student Research Award, \$6000
2018	UBC Science Faculty International Student Scholarship, \$10000
2015	UBC Outstanding International Student Award, \$6000

Selected Service

Reviewer	Neurips 2026, ICML 2025; ICLR 2025; NeurIPS 2024
Organizer	Montreal MLOpt
Organizer	Women@Mila
Volunteer	WiML; ICML 2025, NeurIPS 2023; UAI 2023
Student Reviewer	Graduate Recruiting Committee at Mila 2024; UBC 2022
Volunteer	UBC Computer Science Tri-Mentoring Program, 2018

Teaching Assistant

Sept. 2023 - Dec. 2023	Polytechnique Montreal INF8245E - Machine Learning
Sept. 2021 - May 2023	UBC · CPSC 532M/340: Machine Learning and Data Mining
Jan. 2018 - Apr. 2018	UBC · Math 152: Linear Systems
Sept. 2017 - Apr. 2018	UBC · CPSC 121: Models of Computation
July 2017 - Aug. 2017	UBC · Math 102: Integral Calculus

Selected Presentations

June 2026	Signal frequency Imbalance and Ill-Conditioning <i>Friday Talks @Tuebingen</i>
Nov. 2024	Understanding Adam Requires Better Rotation Dependent Assumptions <i>MTL MLOpt</i>
June 2024	On PI Controllers for Updating Lagrange Multipliers in Constrained Optimization <i>Tsinghua University, Statistical Artificial Intelligence & Learning Group</i>
Apr. 2023	Deep exploration via randomized value function <i>UBC · Reinforcement Learning Reading Group</i>
Oct. 2022	Language Models are Few Shot Learners (GPT-3) <i>UBC · Machine Learning Reading Group</i>
Dec. 2021	Reinforcement Learning and Autonomous Driving <i>UBC · CPSC 340: Machine Learning and Data Mining</i>
Nov. 2021	Probabilistic Topic Modeling <i>UBC · CPSC 503: Computational Linguistics</i>

Work Experience

Jan. 2026 - June 2026	Research Intern <i>OpenEuroLLM</i> - Conducted research on optimization and training dynamics for large language models, resulting in two publications on optimizer behavior and scaling laws. - Developed theoretical analyses and large-scale experiments to characterize optimizer behavior under signal imbalance and identify regimes where adaptive methods outperform SGD. - Designed and executed transformer training experiments to study optimization dynamics, gradient structure, and batch-size scaling.
May 2022 - Dec. 2022	Research Mentor <i>UBC-AWS Cloud Innovation Center</i> - Advised undergraduate students in building machine learning model prototypes in collaboration with AWS
Sept. 2019 - Apr. 2021	Software Engineer <i>Magnitude - Simba Technologies</i> - Developed JDBC driver for Amazon Redshift Database using Java and SQL; Designed test packages and website interface using React.js and HTML/CSS
July 2018 - Aug. 2018	Research Intern <i>Bank of China International, London</i> - Trained regression models for U.S. 10 Year Treasury data and composed financial reports on futures and options predictions

Competitions

Dec. 2018	Putnam Competition, Mathematical Association of America <i>Ranked top 1500 in North America</i>
Sept. 2014	Chinese Physics Olympiad (CPhO) <i>Guangdong Provincial Secondary Prize</i>

Language

Mandarin (native), English (fluent), Cantonese (intermediate), French (beginner)